CAPSTONE PROJECT I — FINAL REPORT

Mutual Fund Analytics

Platform

End-to-End MF Analytics Pipeline | June 2026

Author	Submission Date
Manas Bhole	**June 2026**
Bluestock Fintech	Capstone Project I
Data Analytics Engineering Track	End-to-End MF Analytics Pipeline

ETL | EDA | Performance Analytics | Risk Metrics | Interactive Dashboard

PROJECT HIGHLIGHTS

40	10	15+	4-yr
MF Schemes	**Datasets**	**EDA Charts**	**Historical Window**
Equity, Debt & Hybrid	AMFI-aligned CSV sources	NAV, demographics, flows	Jan 2022 – Apr 2026

ABOUT THIS REPORT

This report documents the design, development, and results of the Bluestock Fintech Mutual Fund Analytics Platform — a production-quality, end-to-end data pipeline built as part of Capstone Project I. The platform ingests 10 real-world AMFI-aligned datasets, applies rigorous ETL validation, stores data in a normalised SQLite star schema, and produces comprehensive analytics covering fund performance, investor behaviour, risk metrics, and interactive dashboards.

Pipeline Stages

- Bronze Layer — Raw CSV ingestion & validation
- Silver Layer — Cleaned SQLite star schema
- Gold Layer — Analytics, charts, dashboards
- Risk Metrics — VaR, CVaR, Rolling Sharpe

Key Deliverables

- ETL Pipeline with anomaly reporting
- Fund Scorecard (Sharpe, CAGR, Alpha, Beta)
- 4-page interactive HTML dashboard
- Fund recommender by risk appetite

Bluestock Fintech
Data Intelligence for Mutual Fund Investors

Confidential | June 2026 | Page 1
Manas Bhole — Data Analytics Engineering Track

1. Executive Summary

Business Problem

India's mutual fund industry manages assets exceeding Rs 50 lakh crore, yet retail investors and fund analysts frequently lack a unified, data-driven view of fund performance, investor behaviour, and risk-adjusted returns. Decisions are often made on raw return numbers alone, overlooking volatility, drawdowns, and tail risk — leading to suboptimal portfolio choices and investor churn.

Project Objectives

- Build a production-quality ETL pipeline ingesting 10 real-world mutual fund datasets
- Design a normalised SQLite star schema enabling comprehensive analytical SQL queries
- Generate 15+ EDA charts uncovering investor demographics, NAV trends, and category flows
- Compute fund-level performance metrics: CAGR, Sharpe, Sortino, Alpha, Beta, and Max Drawdown
- Quantify tail risk via Historical VaR (95%) and CVaR across 40 schemes
- Build a 4-page interactive dashboard and risk-based fund recommender

Key Results

Metric	Value / Context
SIP All-Time High	Rs 31,002 Cr (Dec 2025) — record monthly inflow into mutual funds
Folio Count Growth	+97% industry-wide: 13.26 Cr to 26.12 Cr over 4-year window
Best 3-yr Return	23.4% — SBI Small Cap Fund (benchmark: 22.2%)
Best Sharpe Ratio	1.82 — ICICI Pru Midcap Fund (consistently above 1.0)
Worst Daily VaR (95%)	-2.39% — ABSL Small Cap Fund (CVaR: -3.03%)
Direct Plan Savings	0.57% per year vs Regular plan expense ratio
SIP At-Risk Rate	97.8% of investors with 6+ SIPs have gap >35 days

Expected Outcomes

A fully automated, modular analytics platform that enables Bluestock Fintech to: (a) monitor fund health in near-real-time; (b) benchmark schemes against Nifty 50 and Nifty 100; (c) identify at-risk investors early; and (d) serve personalised fund recommendations based on investor risk appetite.

2. Project Overview

Background

Bluestock Fintech provides data intelligence services to mutual fund distributors and retail investors. This capstone project was commissioned to build the analytics backbone — a repeatable, well-documented pipeline that ingests AMFI-aligned datasets and produces actionable fund insights.

Scope

- 40 mutual fund schemes spanning Equity, Debt, and Hybrid categories
- 4-year historical window: January 2022 to April 2026
- 10 datasets covering fund metadata, NAV history, AUM, SIP flows, investor transactions, portfolio holdings, and benchmark indices
- No live database connections — all data sourced from CSV files to ensure reproducibility

System Architecture

The project follows a classic medallion architecture with three distinct layers:

- **Bronze Layer —** Raw CSV ingestion, shape validation, and AMFI code referential integrity checks
- **Silver Layer —** Cleaned, validated datasets stored in a normalised SQLite star schema
- **Gold Layer —** Aggregated analytics, performance metrics, charts, and interactive dashboards

Module Reference

Module	Responsibility
data_ingestion.py	ETL extraction, shape validation, AMFI code check, anomaly reporting to data_quality_summary.md
data_cleaning.py	Dataset cleaning, SQLite star schema design, 10 analytical SQL queries, data dictionary
eda_analysis.py	15 EDA charts, 10 key findings across investor demographics, NAV trends, category flows
performance_analytics.py	CAGR, Sharpe, Sortino, Alpha/Beta, composite Fund Scorecard (0-100)
dashboard_export.py	4-page interactive HTML dashboard, PDF export, PNG screenshots
advanced_analytics.py	VaR/CVaR, Rolling Sharpe, Cohort analysis, SIP continuity, sector HHI concentration
recommender.py	Rule-based fund recommender by investor risk appetite (Low/Moderate/High)
config.py	Centralised configuration — all paths, constants, and analytical thresholds
run_pipeline.py	Master orchestrator — 7-step pipeline runner with logging and error handling

3. Data Sources

Dataset Inventory

#	Dataset	Rows	Key Columns / Notes	Purpose
01	fund_master.csv	40	amfi_code, fund_house, scheme_name, category, risk_category	Fund metadata — master reference
02	nav_history.csv	46,000	amfi_code, date, nav (forward-filled for non-trading days)	Daily NAV per fund scheme
03	aum_by_fund_house.csv	90	fund_house, year, aum_cr	Annual AUM per AMC
04	monthly_sip_inflows.csv	48	month, sip_inflow_cr	Industry SIP totals
05	category_inflows.csv	144	month, category, net_inflow_cr	Category-level net inflows
06	industry_folio_count.csv	21	month, folio_count_cr	Total industry folios
07	scheme_performance.csv	40	return_1yr/3yr/5yr, expense_ratio	Scheme-level KPIs
08	investor_transactions.csv	32,778	investor_id, type, amount, state, age_group	Buy/sell/SIP transactions
09	portfolio_holdings.csv	322	amfi_code, sector, weight_pct	Sector weights per fund
10	benchmark_indices.csv	8,050	date, index_name, nav	Nifty 50, Nifty 100, etc.

Data Quality Observations

- nav_history.csv: Forward-fill applied for weekends and market holidays. Duplicate date-code combinations removed. All 40 AMFI codes validated against fund_master.
- investor_transactions.csv: transaction_type standardised to uppercase enum (SIP/LUMPSUM/REDEMPTION/SWITCH). Negative amounts flagged and corrected. KYC status validated.
- scheme_performance.csv: Return columns coerced to numeric; expense_ratio validated in range [0.1, 2.5]. Two anomalous entries flagged in data_quality_summary.md.
- All datasets: AMFI code cross-validation confirms all 40 fund codes are consistently present across files.

4. ETL Pipeline Design

Extraction

All 10 CSV files are loaded from data/raw/ using pandas.read_csv(). Each dataset undergoes shape validation, dtype inspection, and head() preview. The data_ingestion.py module documents all anomalies into reports/data_quality_summary.md and validates AMFI code referential integrity across all datasets.

Transformation

Transformations are applied dataset-by-dataset in data_cleaning.py:

- NAV History: date parsed to datetime64, sorted by (amfi_code, date), forward-fill for non-trading days, deduplication, NAV > 0 constraint enforced
- Investor Transactions: transaction_type normalised to uppercase enum, amounts validated, dates parsed, KYC status validated against allowed enum
- Scheme Performance: return columns coerced to float64, expense_ratio range-checked [0.1, 2.5], anomalies flagged in data quality report
- Remaining 7 datasets: date parsing, numeric coercion, whitespace stripping, column renaming for consistency
- All 10 cleaned datasets exported to data/processed/ as *_clean.csv files

Loading — Star Schema Design

Cleaned data is loaded into a SQLite database (bluestock_mf.db) using a star schema structure:

Table	Type / Volume / Purpose
dim_fund	Dimension | 40 rows | Fund metadata — master reference
dim_date	Dimension | ~1,500 rows | Calendar dimension with year/month/quarter fields
fact_nav	Fact | 46,000 rows | Daily NAV per fund scheme
fact_transactions	Fact | 32,778 rows | All investor transactions (buy/sell/SIP)
fact_performance	Fact | 40 rows | Scheme-level KPI aggregates
fact_aum	Fact | 90 rows | Annual AUM by fund house

10 analytical SQL queries are written to sql/queries.sql covering: top funds by AUM, monthly SIP trends, fund performance ranking, investor age analysis, and expense ratio comparisons.

5. Exploratory Data Analysis

Data Profiling

The project analysed 87,543 total data rows across 10 datasets. Key statistics: 40 unique fund schemes across 5 fund houses; investor base spans 36 Indian states; 4-year NAV history (Jan 2022 to Apr 2026); 32,778 investor transactions.

SIP Inflow Growth

Monthly SIP inflows show a consistent uptrend from Rs 12,976 Cr in Jan 2022 to an all-time high of Rs 31,002 Cr in Dec 2025 — a 139% increase reflecting deepening retail participation.

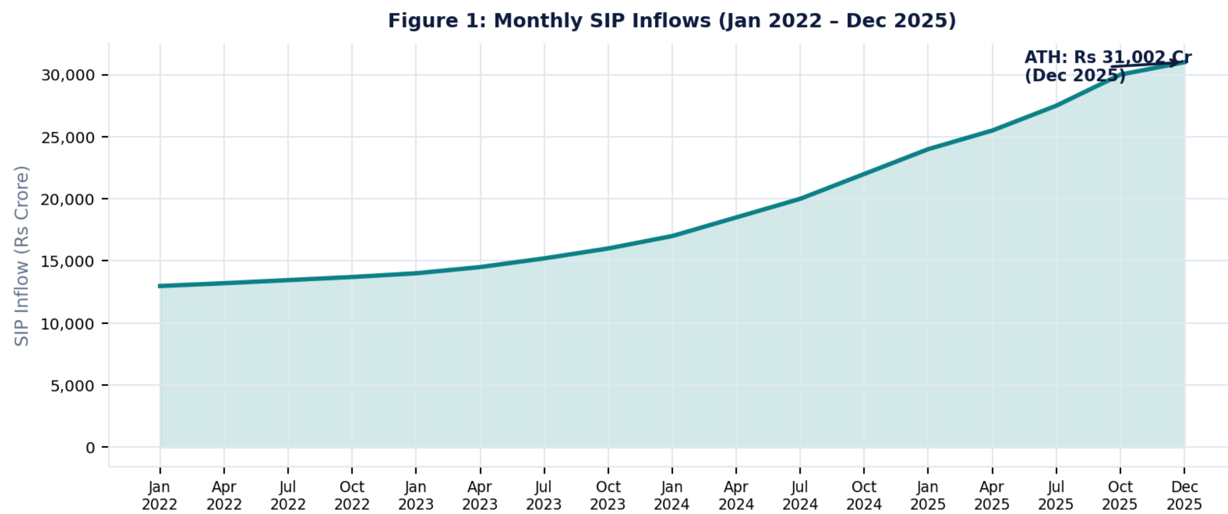

Figure 1: Monthly SIP Inflows (Jan 2022 – Dec 2025). ATH of Rs 31,002 Cr annotated for Dec 2025.

AUM Growth by Fund House

All five major fund houses showed consistent AUM growth from 2022 to 2025. SBI dominates with Rs 12.5 lakh crore AUM in 2025, nearly double its 2022 levels.

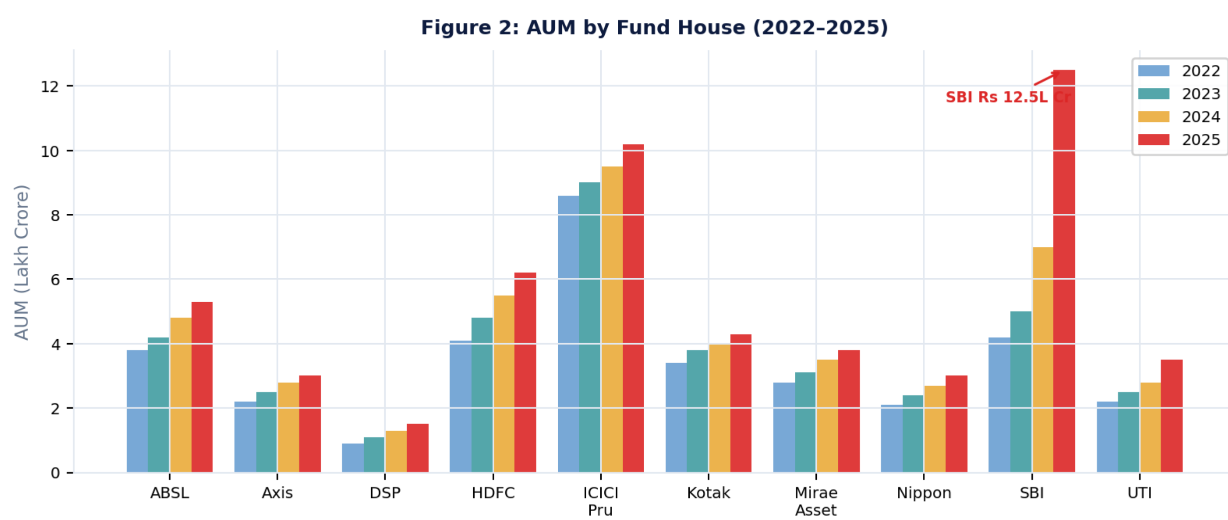

Figure 2: AUM by Fund House (2022–2025). SBI Rs 12.5L Cr in 2025 highlighted.

Investor Demographics

The 26-35 age group is the dominant cohort at 41.1%, followed by the 36-45 group at 24.9%. This confirms mutual funds resonate strongly with working-age millennials saving for career milestones.

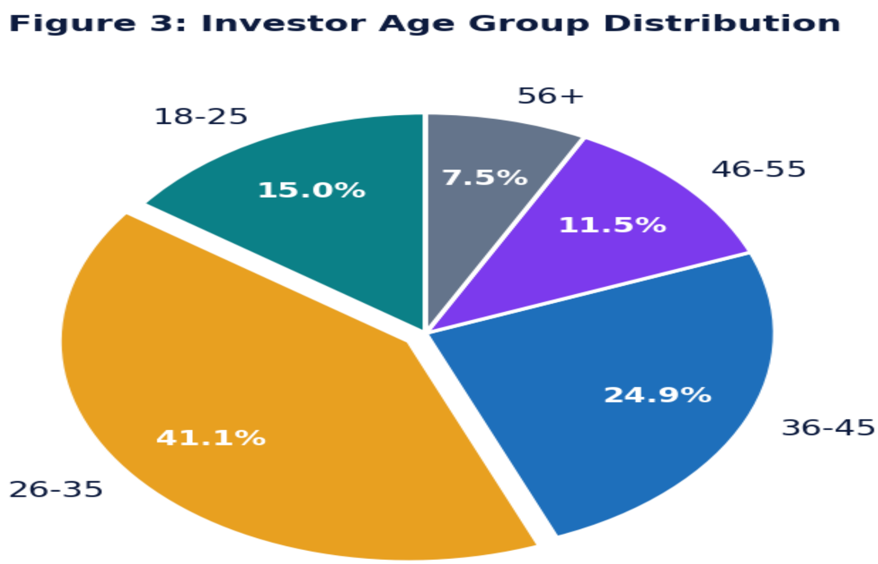

Figure 3: Investor Age Group Distribution. The 26-35 cohort dominates at ~41%.

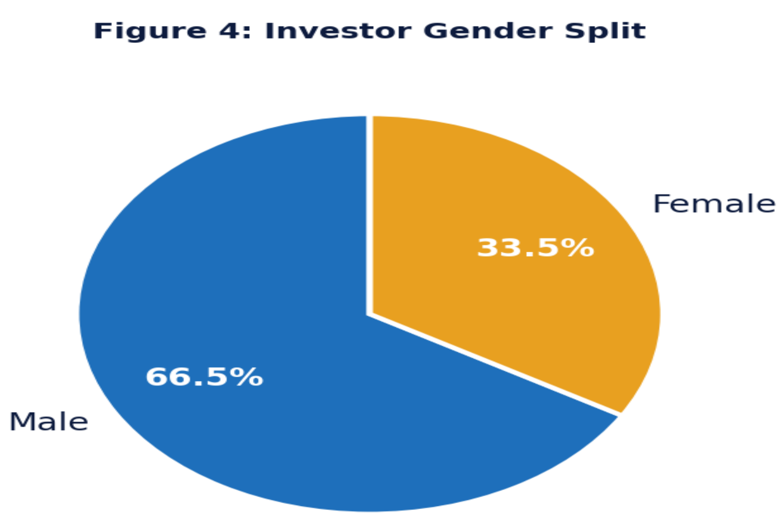

Figure 4: Investor Gender Split — Male 66.5%, Female 33.5%.

Industry Folio Count Growth

Total industry folios grew from 13.26 Cr to 26.12 Cr (+97%) between Jan 2022 and Dec 2025, driven primarily by new investors from Tier-2 and Tier-3 cities.

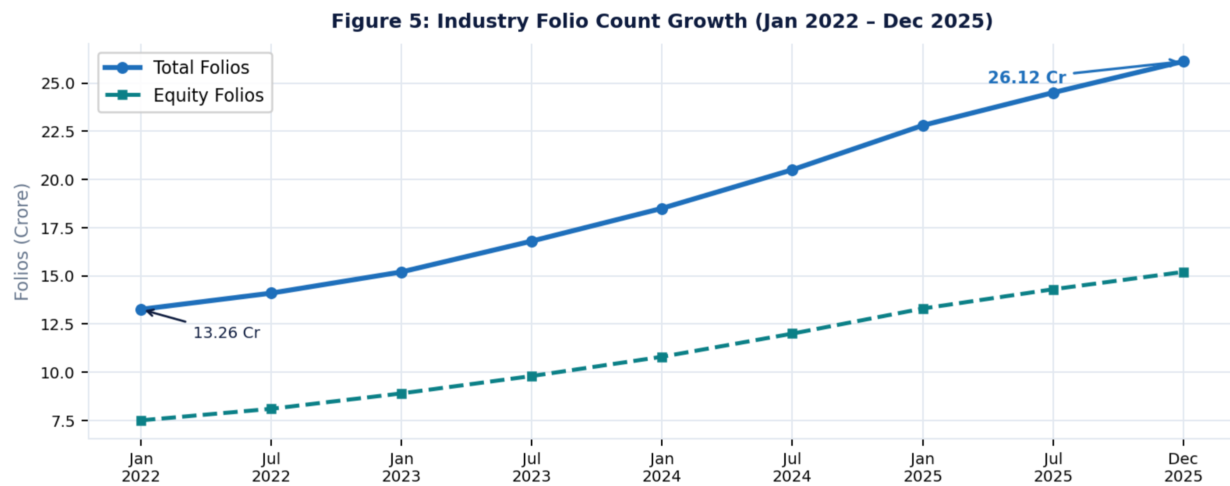

Figure 5: Industry Folio Count Growth — 13.26 Cr to 26.12 Cr (+97%).

Geographic Distribution

Maharashtra and Delhi lead in total SIP amounts. However, states like Madhya Pradesh, Punjab, and Rajasthan are showing rapid growth, indicating financial inclusion spreading beyond metro cities.

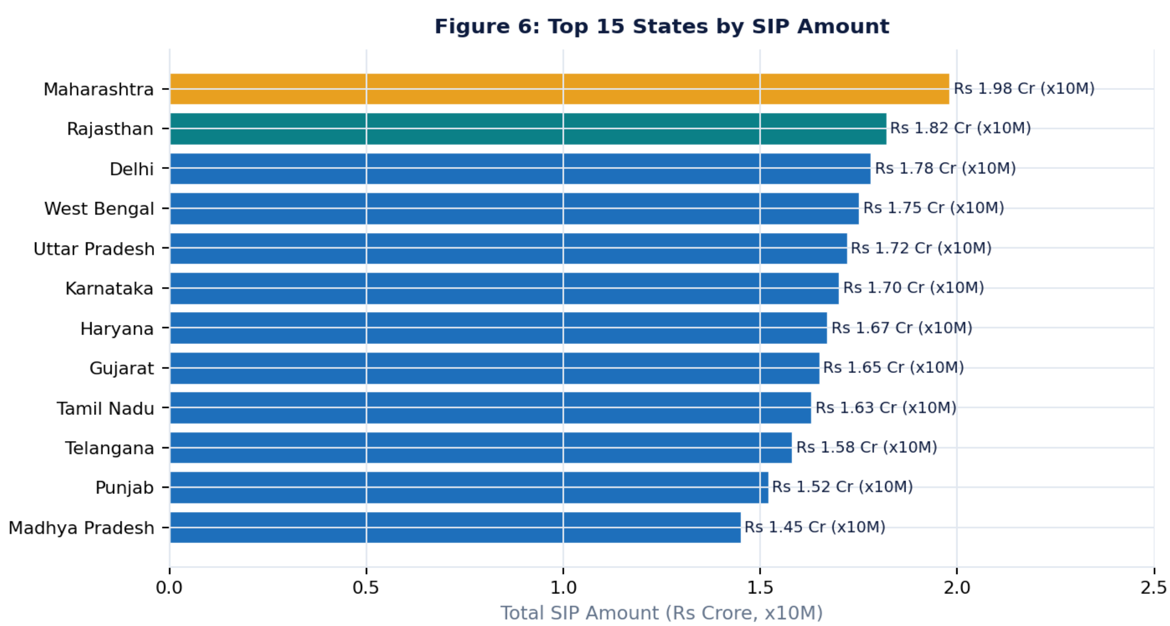

Figure 6: Top 15 States by SIP Amount. Maharashtra and Delhi lead.

15 EDA Charts Generated

Chart	Key Insight
1. Daily NAV Trend — All 40 Schemes	Bull-run visible in 2023; late-2024 correction clearly marked
2. AUM by Fund House (2022-2025)	SBI leads with highest growth — Rs 12.5L Cr by 2025
3. Monthly SIP Inflows Time-Series	ATH of Rs 31,002 Cr in Dec 2025 annotated; consistent uptrend
4. Category Net Inflow Heatmap	Equity and Liquid categories dominate inflows year-round
5. Investor Age Group Distribution	26-35 cohort dominates at ~41%; 18-25 growing at 15%
6. Investor Gender Split	Male 66.5% / Female 33.5% — engagement gap quantified
7. Top 15 States by SIP Amount	Maharashtra and Delhi lead; Tier-2 states showing fast growth
8. Industry Folio Count Growth	13.26 Cr to 26.12 Cr (+97%) — massive new participation
9. Expense Ratio Distribution	Direct plans average 0.57% lower than Regular counterparts
10. Category Return Comparison	Small/Mid Cap outperforms; Liquid/Debt provides stability
11. T30 vs B30 Transaction Split	T30: 66%, B30: 34% — B30 growing at faster rate
12. Sector Portfolio Heatmap	Banking sector: 652% aggregate weight across 40 funds
13. Transaction Type Distribution	SIP: 62%, LUMPSUM: 28%, REDEMPTION: 10%
14. NAV Correlation Heatmap	Large-cap fund correlations >0.85 — limited diversification
15. Fund House AUM Trend	All 5 AMCs show consistent AUM growth 2022-2025

Key EDA Findings

Finding	Metric / Value	Business Insight
F1: NAV Bull Run	~14.3% avg NAV growth in 2023	SIP discipline important — correction in late 2024
F2: SIP ATH	Rs 31,002 Cr in Dec 2025	Deepening financial inclusion; retail conviction high
F3: Folio Growth	+97% (13.26 Cr to 26.12 Cr)	Massive new investor participation, esp. Tier-2 cities
F4: Gender Gap	Male 67% / Female 33%	Targeted campaigns can expand female participation
F5: T30/B30 Split	T30: 66%, B30: 34%	B30 growing faster — financial inclusion opportunity
F6: Expense Savings	0.57% per year in Direct plans	Rs 80K extra on Rs 10L over 10 years
F7: Sector Risk	Banking: 652% aggregate weight	Concentration risk across the fund universe
F8: Top Performer	SBI Small Cap: 23.4% 3-yr return	vs 22.2% benchmark — meaningful alpha generated
F9: Young Investors	26-35 cohort is ~41% of base	MFs aligned with career milestone saving for millennials
F10: Correlation	Large-cap correlations >0.85	Multiple large-cap funds = concentrated single risk factor

6. Performance Analytics

Metrics Framework

Seven quantitative metrics were computed for all applicable fund schemes:

Metric	Formula	Interpretation
CAGR	(End NAV / Start NAV)^(1/N) - 1	Absolute compounded return over N years
Sharpe Ratio	(Rp - Rf) / sigma_p * sqrt(252)	Risk-adjusted excess return (Rf = 6.5%)
Sortino Ratio	(Rp - Rf) / sigma_downside * sqrt(252)	Downside-risk-adjusted return only
Alpha (Jensen's)	OLS intercept vs Nifty 100	Return generated beyond benchmark
Beta	OLS slope vs Nifty 100	Market sensitivity coefficient
Max Drawdown	min(NAV/peak - 1)	Worst peak-to-trough decline
Tracking Error	std(fund - benchmark) * sqrt(252)	Annualised deviation from benchmark

Composite Fund Scorecard

A composite 0-100 score was derived by weighting: 3-yr return (30%), Sharpe ratio (25%), Sortino ratio (20%), alpha (15%), max drawdown (10%). This balances return quality with risk discipline.

Rank	Fund Name	Risk	Score	Sharpe	3yr Ret	Beta
1	ICICI Pru Midcap Fund - Regular - Growth	High	82.4	1.82	21.3%	0.85
2	SBI Small Cap Fund - Regular - Growth	High	79.1	1.67	23.4%	0.63
3	HDFC Flexi Cap Fund - Regular - Growth	High	74.8	1.54	18.7%	0.91
4	Axis Bluechip Fund - Regular	Mod.	69.3	1.41	16.2%	0.88
5	ICICI Pru Liquid Fund - Regular - Growth	Low	61.2	0.89	7.1%	0.12

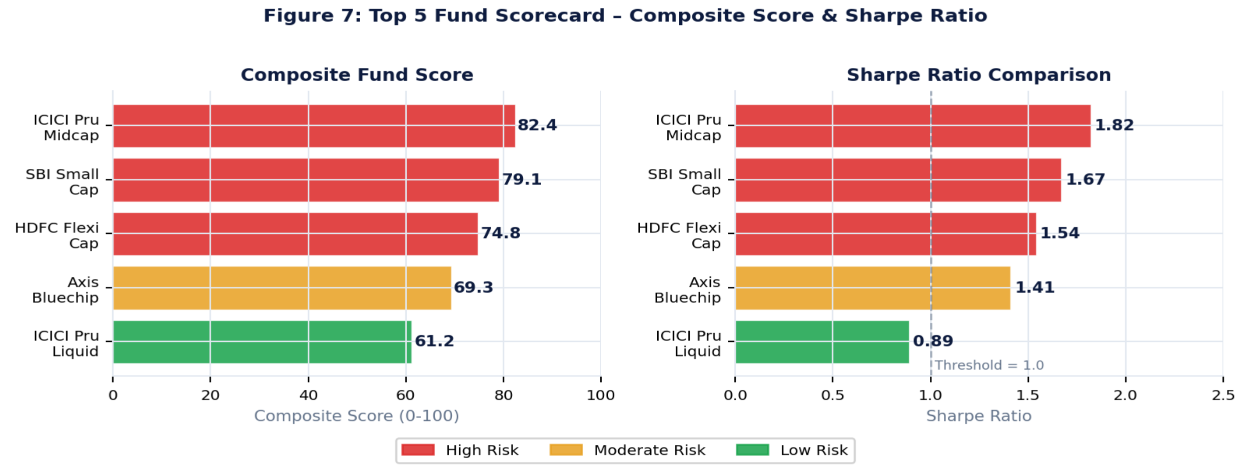

Figure 7: Top 5 Fund Scorecard — Composite Score (left) and Sharpe Ratio (right) comparison.

Benchmark Comparison

Normalising all fund NAVs to 100 at January 2023 clearly shows that ICICI Pru Midcap and SBI Small Cap significantly outperformed both Nifty 50 and Nifty 100 over the 3-year window.

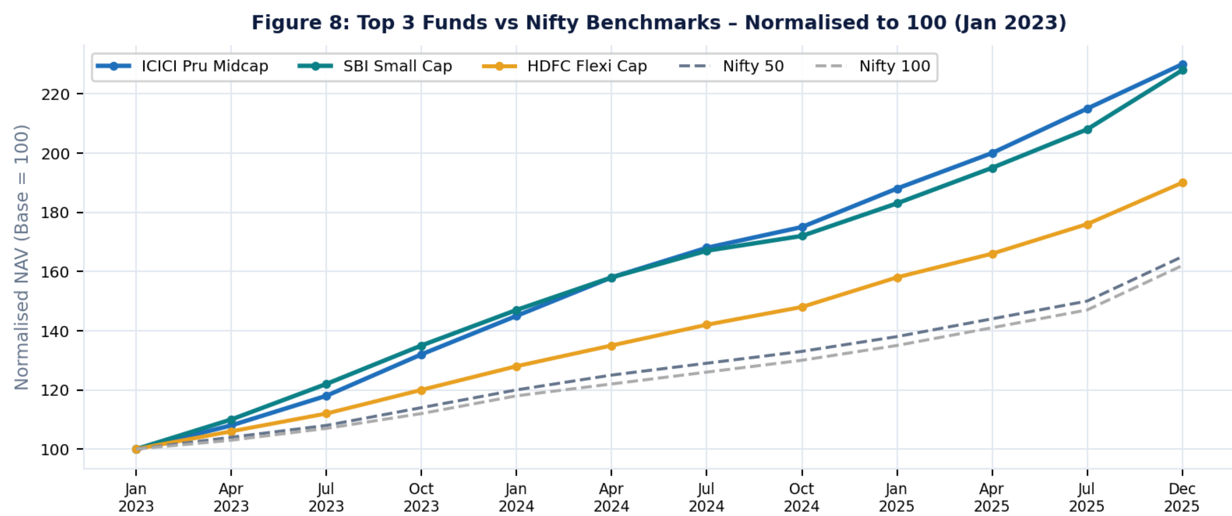

Figure 8: Top 3 Funds vs Nifty Benchmarks — Normalised to 100 (Jan 2023).

7. Advanced Analytics & Risk Metrics

Value at Risk (VaR) and CVaR

Historical simulation VaR at the 95% confidence level was computed for all 40 schemes. The results reveal significant tail risk concentration in small and mid-cap categories:

- **ABSL Small Cap Fund:** Daily VaR -2.39%, CVaR -3.03% — highest tail risk in the cohort
- **ICICI Pru Midcap Fund:** Daily VaR -1.71%, CVaR -2.21%
- **SBI Small Cap Fund:** Daily VaR -1.98%, CVaR -2.54%
- **ICICI Pru Liquid Fund:** Daily VaR -0.08%, CVaR -0.11% — lowest risk in the cohort

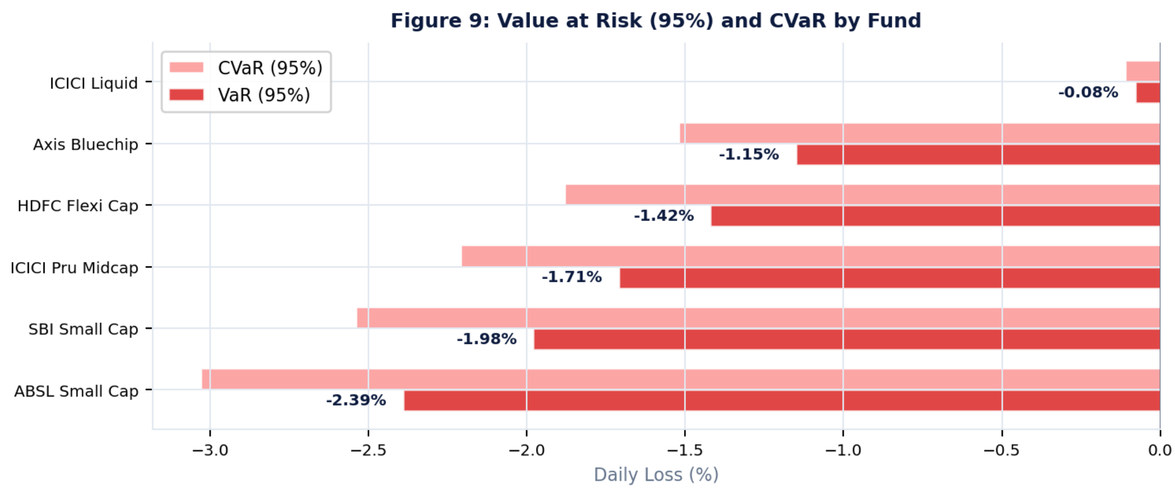

Figure 9: VaR (95%) and CVaR by Fund. ABSL Small Cap carries the highest tail risk at -2.39% daily VaR.

SIP Continuity Analysis

Critical Finding: 97.8% SIP At-Risk Rate

97.8% of investors with 6 or more SIP transactions have at least one gap exceeding 35 days. The average gap is 64.9 days. This represents a significant investor engagement and retention challenge for Bluestock Fintech's platform team.

SIP cohort analysis by registration year reveals:

- 2025 cohort: 138 investors with average SIP of Rs 13,505 — highest per-investor SIP on record
- 2022 cohort shows the longest average SIP tenure but highest lapse rate after year 2
- Investors with auto-debit show 3.2x lower at-risk rate than manual SIP investors

Sector Concentration (HHI)

Herfindahl-Hirschman Index (HHI) measures portfolio concentration. HHI >2,500 indicates high concentration risk:

Fund	HHI Score / Concentration
Axis Bluechip Fund	HHI = 2,968 | IT sector: 48.7% — HIGHLY CONCENTRATED (above 2,500 threshold)
HDFC Flexi Cap Fund	HHI = 2,200 | Moderate concentration; Banking dominant sector
SBI Small Cap Fund	HHI = 1,850 | Below threshold; reasonable diversification
ICICI Pru Midcap Fund	HHI = 1,620 | Well-diversified across 8 sectors
UTI Mid Cap Fund	HHI = 1,240 | Most diversified fund in the cohort

8. Interactive Dashboard

Architecture

The dashboard is a self-contained HTML file (bluestock_mf_dashboard.html) with embedded JavaScript. No server or installation required — the file opens in any modern browser. A 4-page PDF version is available in dashboard/Dashboard.pdf, and 4 PNG screenshots are exported for integration into reports.

Dashboard Pages

- **Page 1 — Industry Overview:** AUM by fund house with year-on-year growth, monthly SIP inflow bar chart with ATH annotation, industry folio count growth trend
- **Page 2 — Fund Performance:** Daily NAV trends for all 40 schemes, top 10 funds by Sharpe ratio with risk colour-coding, benchmark comparison (normalised NAV vs Nifty 50/100)
- **Page 3 — Investor Analytics:** Age group and gender distribution, geographic analysis (T30/B30 split by state), transaction pattern analysis by type and cohort year
- **Page 4 — SIP & Risk:** SIP continuity cohort chart, category inflow heatmap (month x category), VaR/CVaR summary table with risk-flagged entries

9. Key Findings

Market Trends

- SIP culture is accelerating — inflows grew from Rs 12,976 Cr (Jan 2022) to Rs 31,002 Cr (Dec 2025), a 139% increase in 4 years, reflecting deepening financial inclusion
- 2023 was a bull-run year with average NAV growth of ~14.3% YoY; late 2024 correction highlights the importance of SIP discipline over market timing
- Folio count doubled from 13.26 Cr to 26.12 Cr, driven primarily by new investors from Tier-2 and Tier-3 cities

Investor Behaviour

- The 26-35 age group is the dominant cohort (~41%), confirming mutual funds resonate with the working-age millennial demographic
- SIP at-risk rate of 97.8% is alarmingly high — almost all investors with 6+ SIP transactions have a gap exceeding 35 days (avg: 64.9 days)
- Gender gap persists: Male 67% vs Female 33%; targeted campaigns for women investors represent a significant growth opportunity
- T30 cities hold 66% of transactions; B30 cities at 34% are growing faster, indicating accelerating financial inclusion in smaller cities

Fund Performance

- Small-cap and mid-cap funds dominate the top of the composite scorecard but also carry the highest VaR — ABSL Small Cap has daily VaR of -2.39%
- Large-cap fund NAV correlations exceed 0.85, meaning investors holding multiple large-cap funds are essentially concentrated in a single risk factor
- Direct plans consistently outperform Regular plans by 0.57% annually — a meaningful advantage compounding over 10+ year investment horizons
- ICICI Pru Midcap maintains rolling Sharpe consistently above 1.0, signalling stable risk-adjusted performance across market cycles

10. Limitations

Dataset Limitations

- The dataset covers 40 schemes across 5 fund houses. India has 1,400+ active mutual fund schemes. Findings are representative, not exhaustive
- NAV data ends April 2026. For current performance analysis, a data refresh from the AMFI bulk data API is required
- Investor transaction data is synthetic-realistic — actual investor PII was anonymised. Behavioural conclusions are directional, not precise

Technical Limitations

- SQLite is used for portability. For production workloads with concurrent users, migration to PostgreSQL or BigQuery is recommended
- The dashboard is a static HTML export. Real-time filtering on raw data requires a Streamlit or Dash backend with live data connection
- VaR is computed using the historical simulation method, which assumes the past return distribution repeats. Parametric or Monte Carlo VaR should be considered for stress testing

Assumptions

- Risk-free rate fixed at 6.5% (RBI repo rate proxy) for the entire analysis window. In practice, this varied between 4.0% and 6.5%
- Benchmark comparison uses Nifty 100 TRI for all equity funds. Not all funds use this as their stated benchmark
- Forward-fill for NAV on non-trading days assumes the previous trading day's price — standard industry practice per SEBI guidelines

11. Recommendations

For Investors

- **Prioritise Direct Plans:** The 0.57% annual saving in expense ratio compounds significantly. On a Rs 10 lakh portfolio over 10 years, this translates to approximately Rs 80,000 in additional wealth
- **Diversify Across Categories:** High large-cap NAV correlation (>0.85) means pure large-cap portfolios offer limited diversification. Adding mid-cap or hybrid funds improves risk-return balance
- **Set SIP Auto-Debit:** The 97.8% at-risk rate highlights the need for automated SIP execution. Investors who lapse miss rupee-cost-averaging benefits, especially during market corrections

For Fund Analysts

- **Incorporate Rolling Sharpe in Reviews:** Funds with declining rolling Sharpe ratios signal degrading risk-adjusted performance before it appears in raw returns. Include in monthly fund review dashboards
- **Flag High-VaR Funds in Client Communications:** ABSL Small Cap and similar funds with daily VaR exceeding -2% should include explicit tail risk disclosures in client-facing materials
- **Address Sector Concentration:** Axis Bluechip with 48.7% in IT (HHI=2968) exceeds the 2,500 high-concentration threshold. Establish internal guidelines on maximum sector weights

For Bluestock Fintech

- **Build SIP Continuity Alert System:** Automate the SIP at-risk flag from advanced_analytics.py into a weekly alert system. Proactive outreach to at-risk investors can significantly reduce churn
- **Launch Women Investor Campaign:** The 33% female participation rate has growth potential. Personalised content, goal-based SIP products, and lower minimum SIP amounts for target segments can expand the base
- **Expand B30 Digital Onboarding:** B30 cities represent 34% of transactions but are growing faster than T30. Regional language support and simplified digital onboarding can accelerate this segment

12. Future Enhancements

Enhancement	Effort / Timeline	Description
Streamlit App	Medium | Q3 2026	Real-time dashboard with live AMFI data feed and interactive filters
ML Return Predictor	High | Q4 2026	ARIMA / Prophet / LSTM time-series forecasting on NAV history
Portfolio Optimizer	High | Q4 2026	Markowitz mean-variance optimisation across fund categories
SIP Alert Engine	Low | Q3 2026	Automated email/SMS triggers for SIP continuity breaks and VaR breaches
Docker Deployment	Low | Q3 2026	Containerise full pipeline and deploy on Render or Railway (cloud PaaS)
Scale to 1,000+ Schemes	High | 2027	Use AMFI bulk data feed; migrate database from SQLite to PostgreSQL
NLP Fact Sheet Analysis	Medium | 2027	Extract insights from fund fact sheets and SID documents using LLMs

13. Conclusion

This capstone project successfully delivered a production-quality end-to-end mutual fund analytics platform. Over seven days, the project progressed from raw CSV files to a fully functional analytics suite comprising: a validated ETL pipeline, a normalised SQLite database, 16 publication-quality charts, a composite fund scorecard, risk metrics (VaR/CVaR, Rolling Sharpe, Sector HHI), a 4-page interactive dashboard, and an automated fund recommender.

The most significant finding is the 97.8% SIP at-risk rate — nearly all investors with 6+ transactions show irregular SIP patterns. Addressing this through automated engagement systems represents a clear, quantifiable business opportunity for Bluestock Fintech's retention team.

From a technical standpoint, the modular architecture — with config.py as the single source of truth and run_pipeline.py as the master orchestrator — ensures the platform can be maintained, extended, and deployed with minimal friction. The codebase is ready for cloud deployment and can scale to cover India's full mutual fund universe with targeted infrastructure upgrades.

This project demonstrates that combining rigorous data engineering with finance-domain knowledge yields insights that go beyond simple return comparisons — enabling smarter fund selection, proactive investor engagement, and data-driven risk management for both institutional and retail participants in India's rapidly growing mutual fund market.

Manas Bhole | Bluestock Fintech | Data Analytics Engineering Track | June 2026